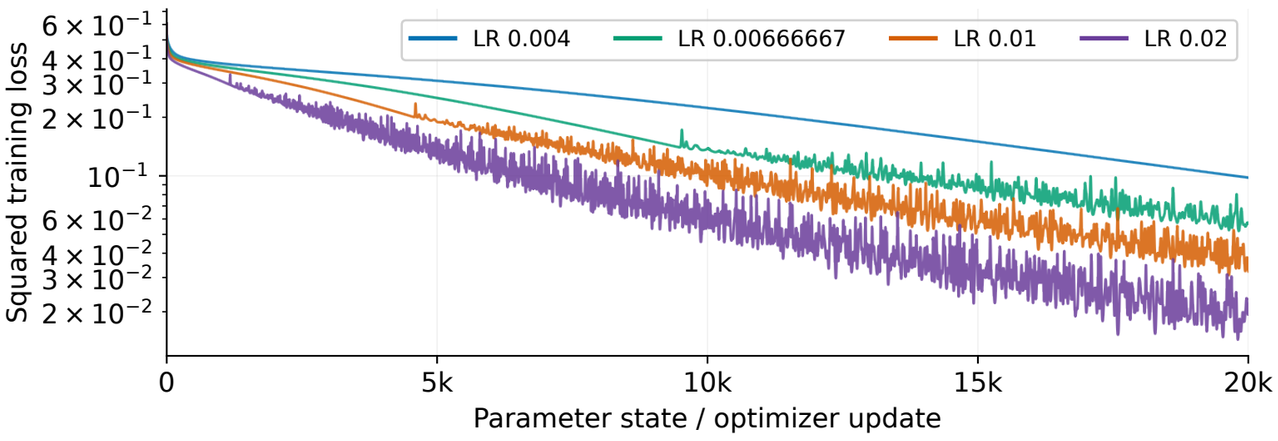
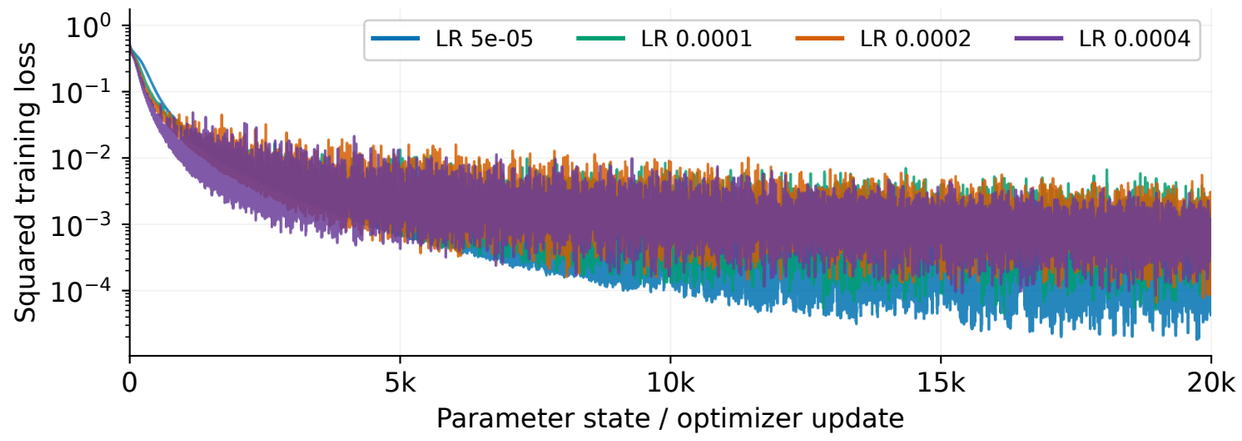
CIFAR-10 full-batch learning-rate sweep | GD and Adam
5,000 training images | tanh 200-200 MLP | one frozen seed | 20,000 updates

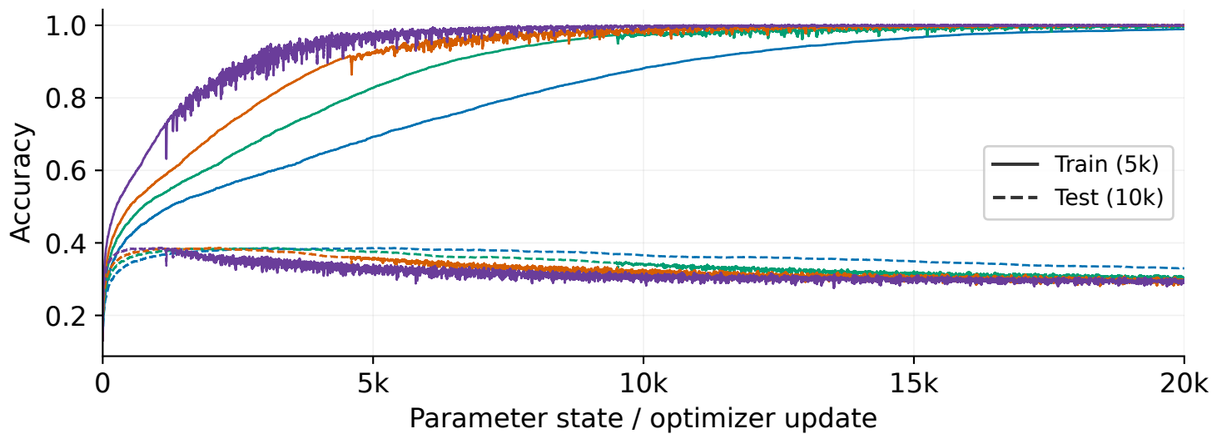
(a) GD: unsmoothed training loss



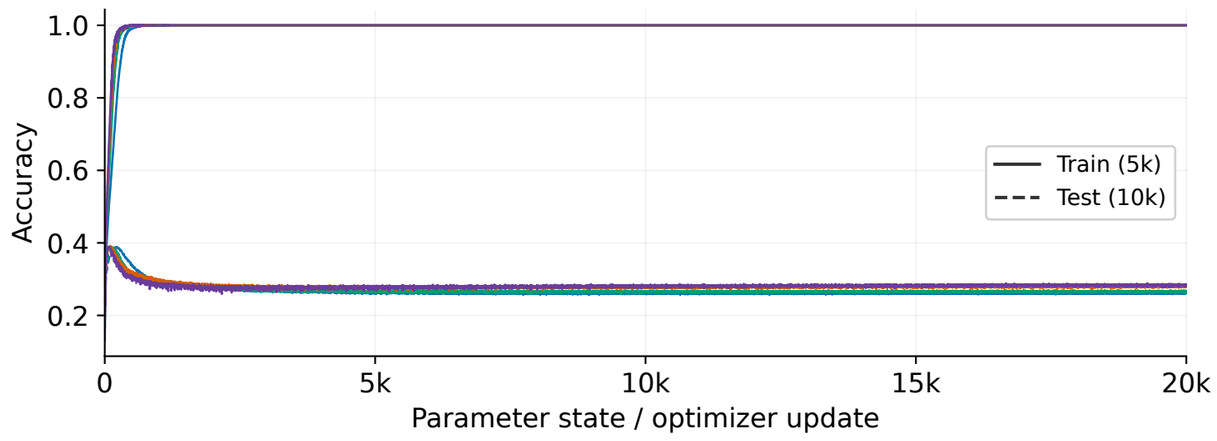
(b) Adam: unsmoothed training loss



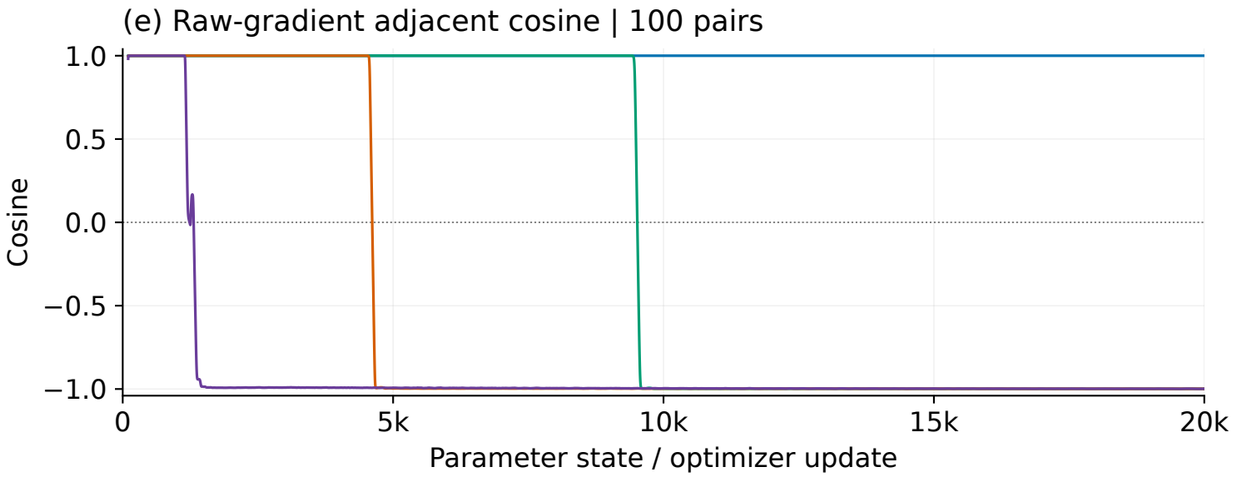
(c) Training and test accuracy



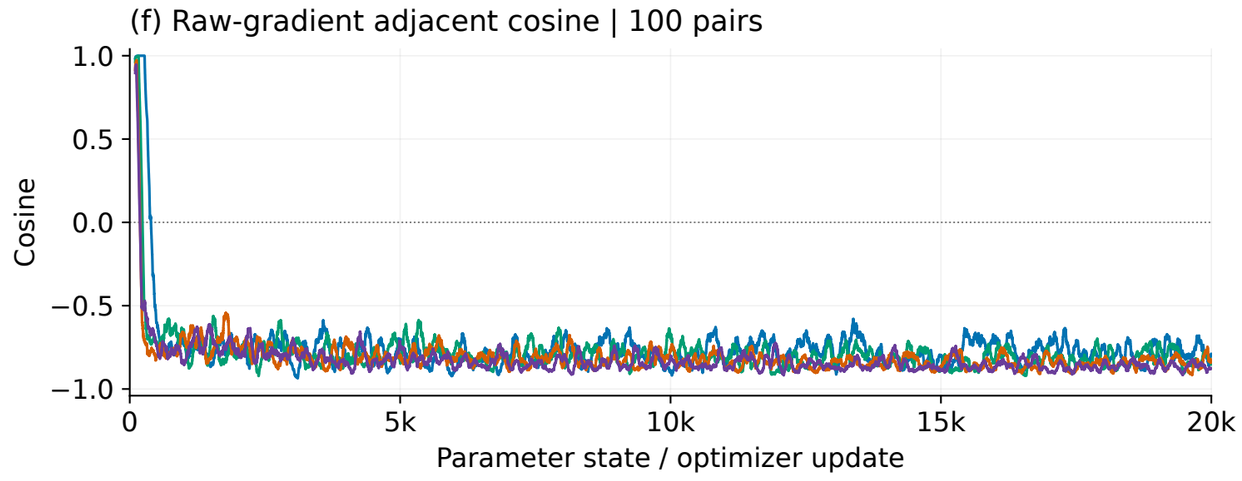
(d) Training and test accuracy



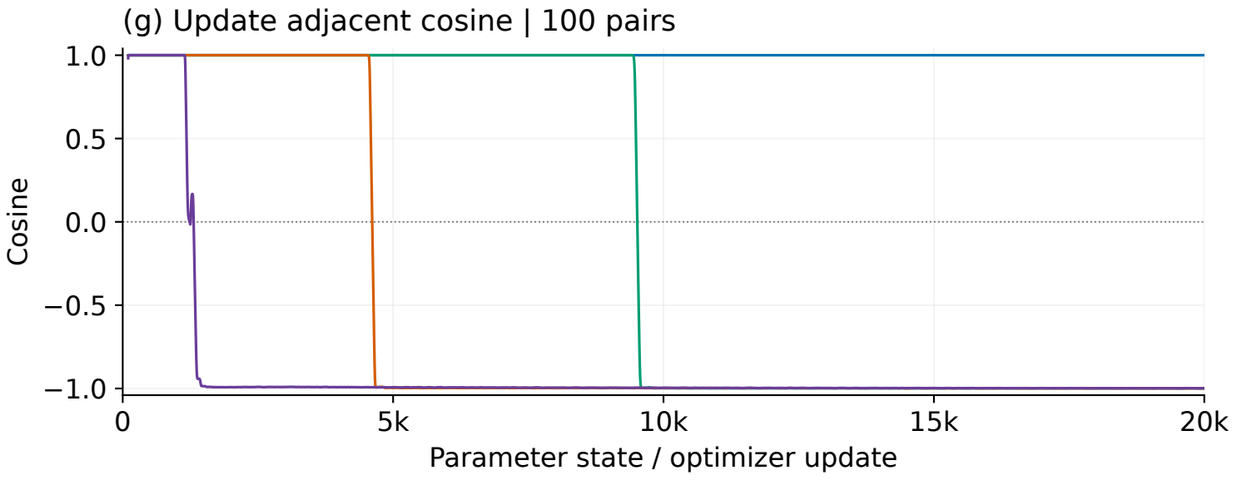
(e) Raw-gradient adjacent cosine | 100 pairs



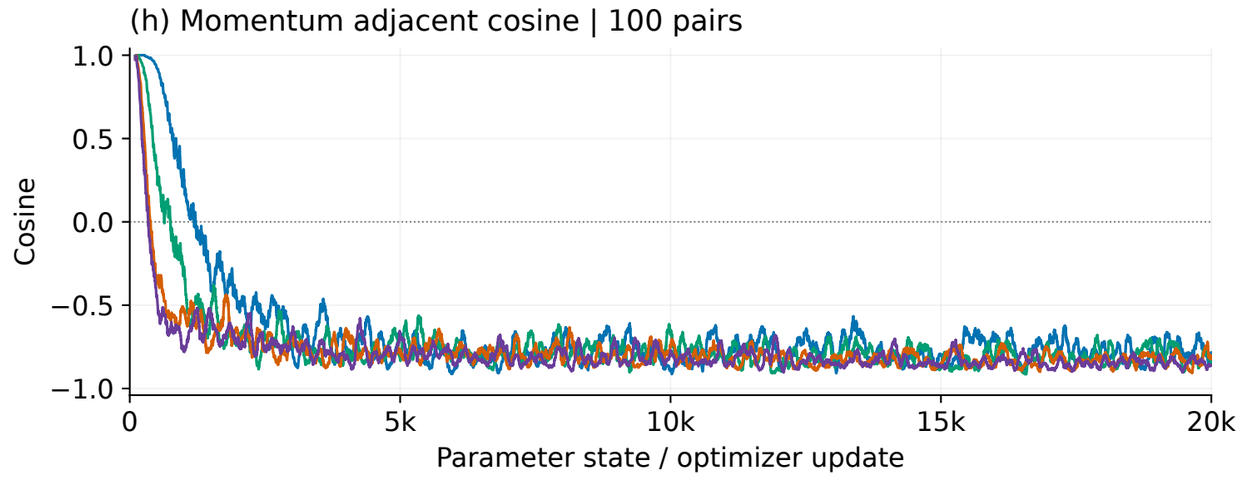
(f) Raw-gradient adjacent cosine | 100 pairs



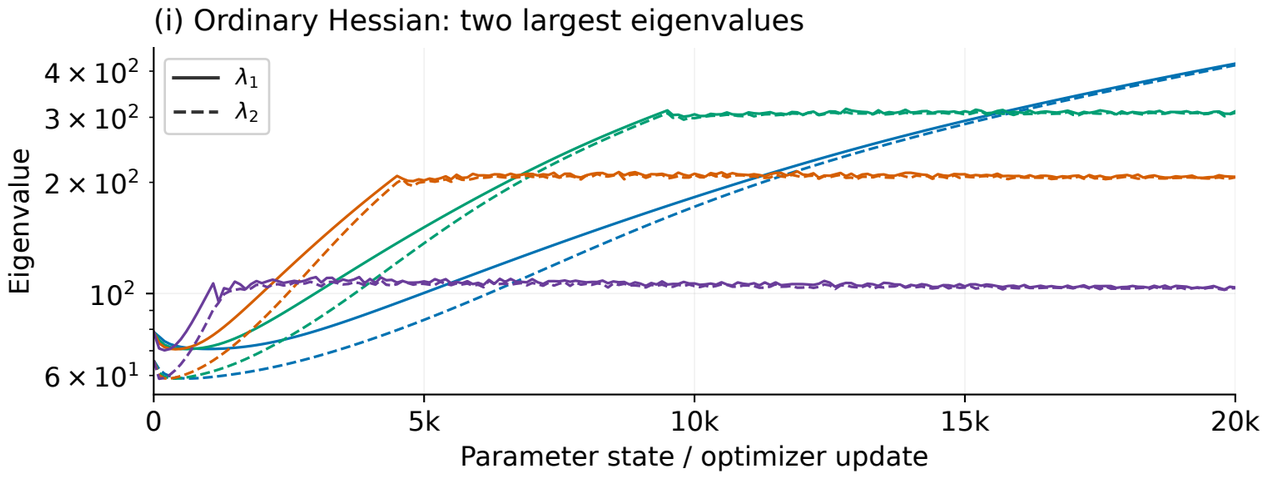
(g) Update adjacent cosine | 100 pairs



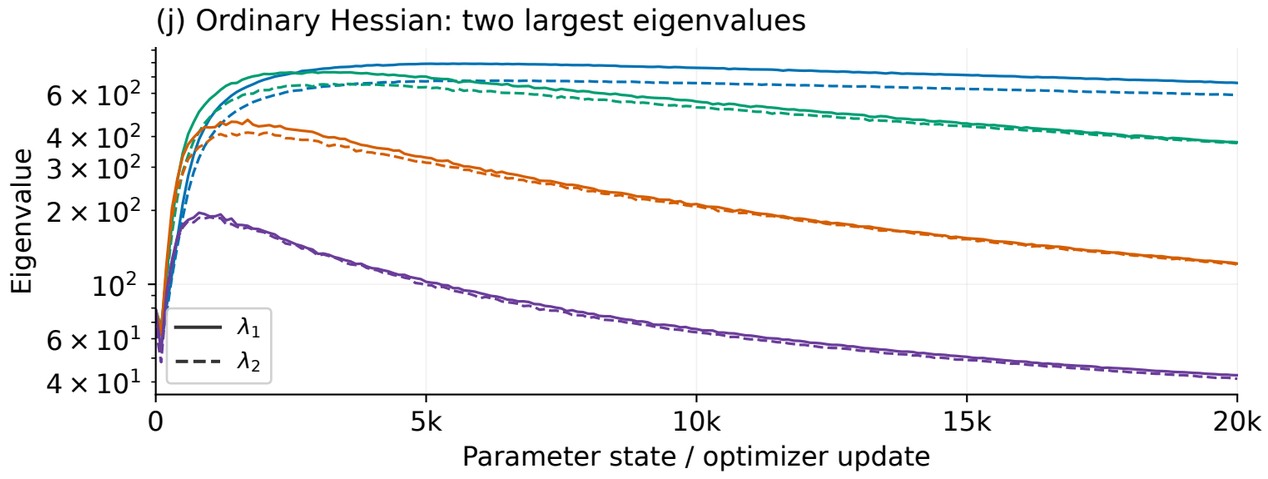
(h) Momentum adjacent cosine | 100 pairs



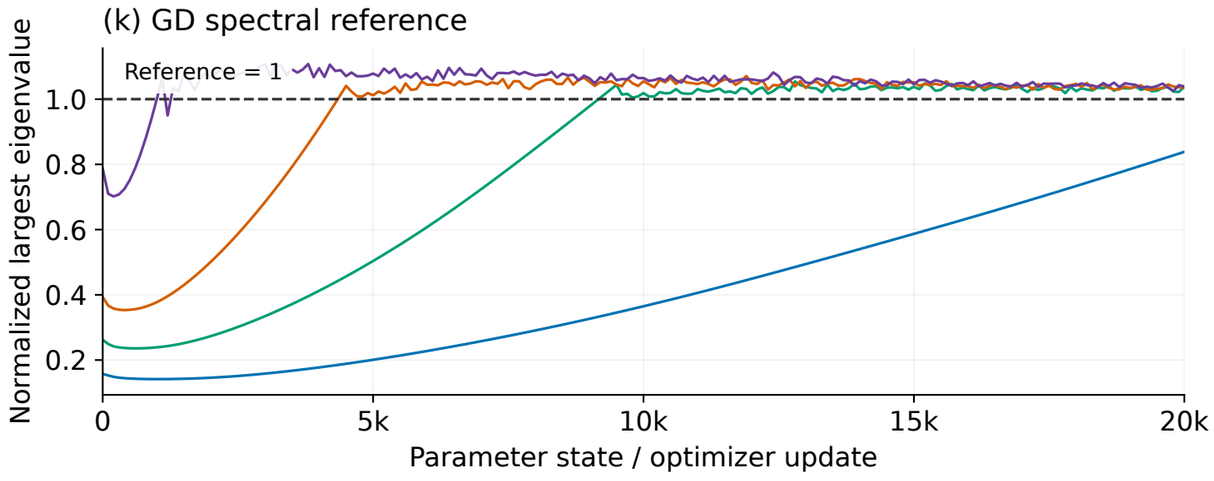
(i) Ordinary Hessian: two largest eigenvalues



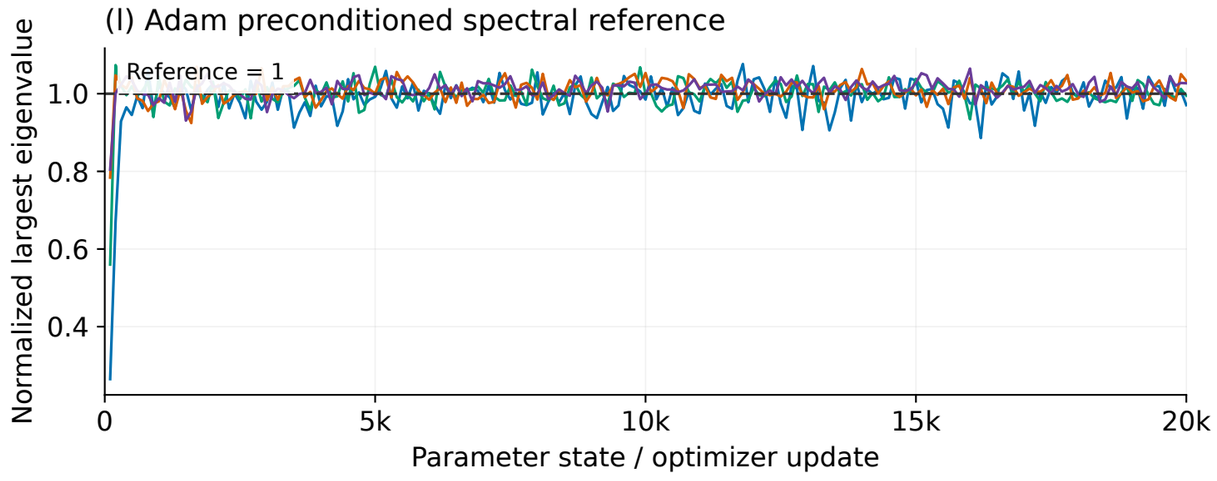
(j) Ordinary Hessian: two largest eigenvalues



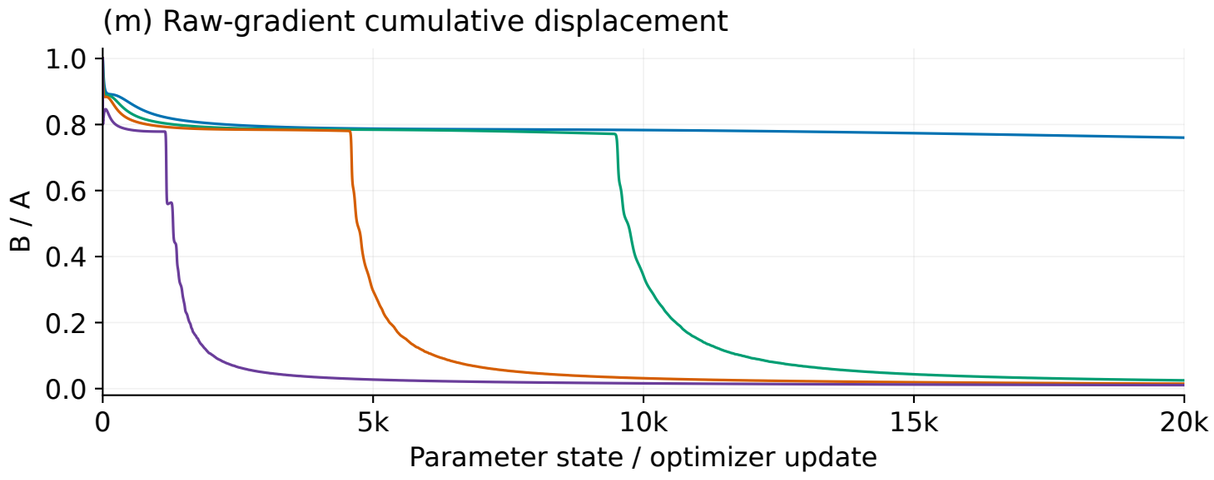
(k) GD spectral reference



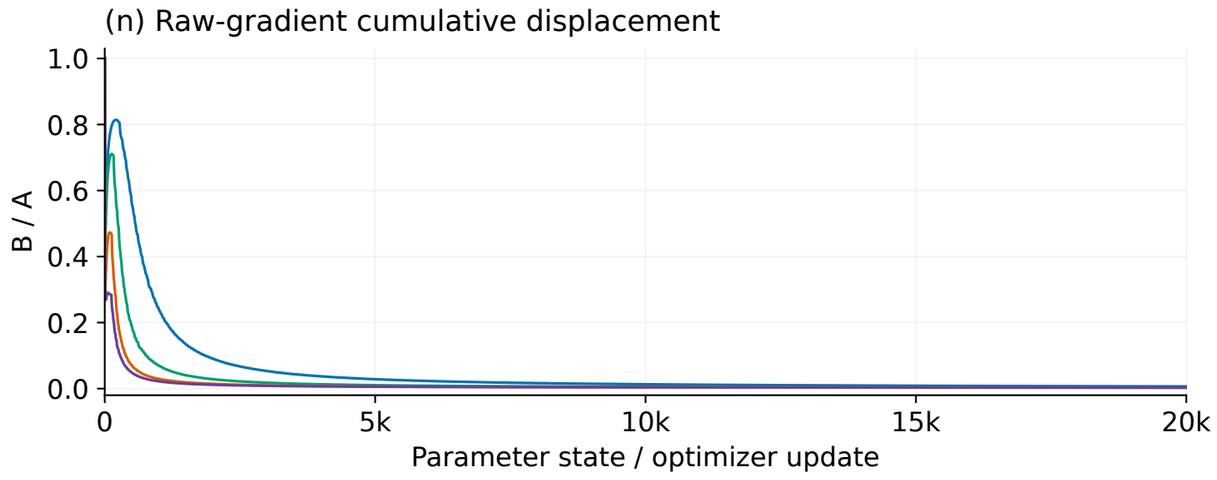
(l) Adam preconditioned spectral reference



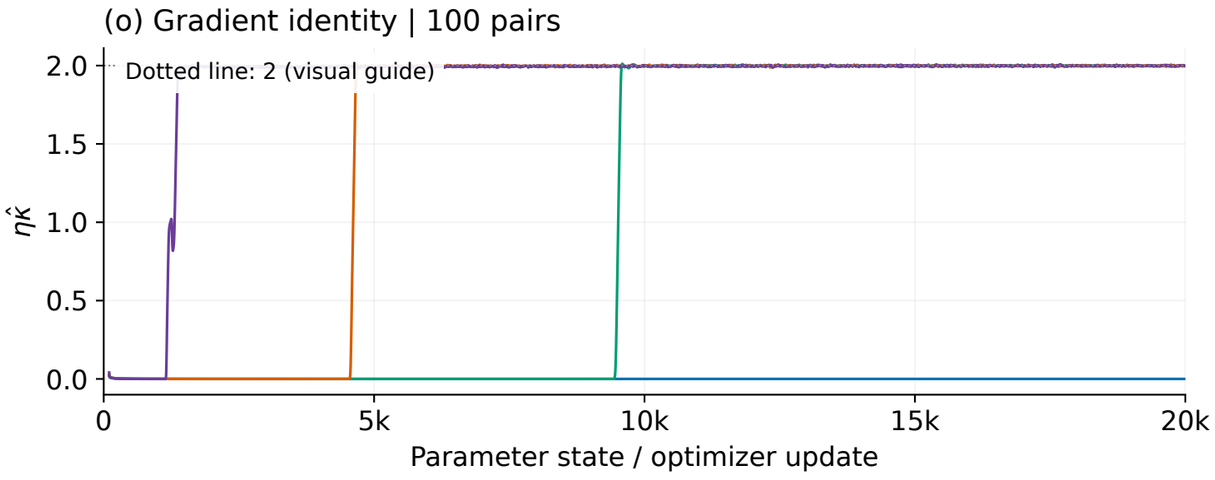
(m) Raw-gradient cumulative displacement



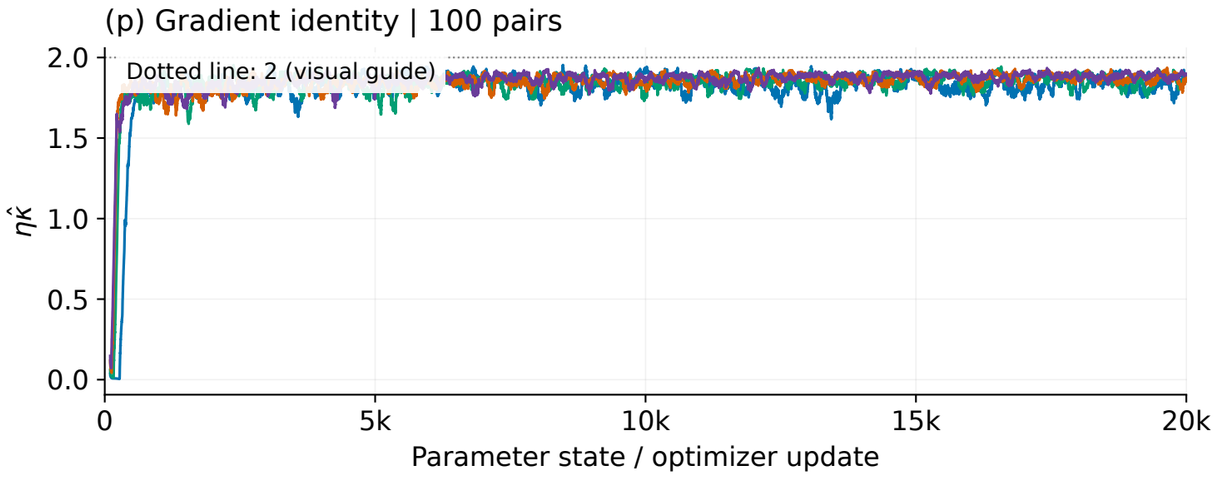
(n) Raw-gradient cumulative displacement



(o) Gradient identity | 100 pairs



(p) Gradient identity | 100 pairs



Train and gradient state t-1; moments and displacement at update t; test and Hessian at their labeled state.
Adam: P=(1-beta1^t)(sqrt(vhat)+epsilon); 38/eta is a frozen-preconditioner local reference. Kappa is a proxy, not an Adam curvature identity.